

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Spring 2026
Parallel & Distributed Computing (CS 429)
Assignment # 3

Semester: 8th
Announced Date: 14-May-26
Total Marks: 4
Instructor Name: Dr. Zaheen Fatima

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Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 3	Analyze concepts of interconnection networks, performance evaluation, and message-passing (MPI) to develop efficient parallel and distributed programs..	PLO_3 (Problem Analysis)	C4 (Analyzing)

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Secrion: B

Question: A government organization is developing a Smart City Traffic Management System to monitor and control traffic signals across multiple cities in real time. The system collects data from thousands of IoT sensors, CCTV cameras, GPS-enabled vehicles, and RFID toll stations distributed throughout the road network. Due to the huge amount of incoming traffic data, a single computer cannot process the information efficiently. Therefore, the development team decides to use a **parallel and distributed computing environment** consisting of multiple interconnected processors and distributed servers. The servers communicate using **Message Passing Interface (MPI)** to exchange traffic information such as vehicle count, congestion level, accident reports, and signal timing updates.

The team is evaluating different **interconnection network topologies** including:

- Bus topology
- Ring topology
- Mesh topology
- Hypercube topology

The engineers also want to measure system performance by analyzing:

- Speedup
- Efficiency
- Scalability
- Communication overhead
- Latency and throughput

The system must:

- Process traffic data in parallel
- Share information among distributed nodes
- Reduce communication delay
- Handle increasing numbers of vehicles and sensors efficiently

Questions

Part A — Interconnection Networks

1. Define an interconnection network in parallel and distributed systems.

2. Compare the following interconnection topologies in terms of:
 - Communication speed
 - Scalability
 - Reliability
 - Cost
 - a) Bus topology
 - b) Ring topology
 - c) Mesh topology
 - d) Hypercube topology
3. Which topology would be most suitable for the Smart City Traffic Management System? Justify your answer.

Part B — Performance Evaluation

The sequential execution time of traffic data processing is:

$T_s = 120$ seconds

$T_p = 120$ seconds

Number of Processors: 8

4. Calculate the Speedup of the parallel system.
5. Calculate the Efficiency of the system.
6. If the number of traffic sensors doubles in the future, explain how scalability affects system performance.
7. Explain how communication overhead can reduce the performance of distributed systems.

Part C — MPI (Message Passing Interface)

8. What is MPI and why is it important in distributed computing?
9. Explain the role of the following MPI functions:
 - `MPI_Init()`
 - `MPI_Send()`
 - `MPI_Recv()`
 - `MPI_Finalize()`
10. Write a simple MPI-based algorithm or pseudocode showing how two processors exchange traffic congestion data.

Part D — Critical Analysis

11. Suppose one server fails during traffic processing:
 - a. How can distributed systems maintain fault tolerance?
 - b. Which interconnection topology would provide better reliability in this case?
12. Suggest two techniques to optimize communication between distributed nodes in MPI-based systems.

Part A – Interconnection Networks

1. Define an Interconnection Network in Parallel and Distributed Systems.

An **interconnection network** is a communication system that connects multiple processors, computers, or servers in a parallel or distributed computing environment. It enables these nodes to exchange data, coordinate tasks, and work together efficiently while executing applications. The performance of a distributed system depends heavily on the design of its interconnection network because it affects communication speed, reliability, scalability, and overall system efficiency.

In a Smart City Traffic Management System, the interconnection network allows distributed servers to share information such as vehicle counts, congestion levels, accident reports, and traffic signal updates in real time.

2. Compare the Following Interconnection Topologies

Feature	Bus Topology	Ring Topology	Mesh Topology	Hypercube Topology
Communication Speed	Low because all devices share one communication channel	Moderate as data passes through intermediate nodes	High due to multiple direct communication paths	Very high because communication requires fewer hops
Scalability	Poor; adding more nodes reduces performance	Moderate; delay increases with more nodes	Excellent; supports large expansion	Excellent; grows efficiently with dimensions
Reliability	Low; failure of the main bus can stop the entire network	Moderate; failure of one node may disrupt communication	High; multiple alternative paths provide redundancy	Very high due to multiple redundant connections
Cost	Low implementation cost	Moderate cost	High because many links are required	High due to complex structure
Fault Tolerance	Poor	Moderate	Excellent	Excellent
Suitability for Large Systems	Not suitable	Limited suitability	Highly suitable	Highly suitable

3. Which Topology is Most Suitable for the Smart City Traffic Management System? Justify Your Answer.

The **Mesh Topology** is the most suitable choice for the Smart City Traffic Management System.

Justification

- It provides multiple communication paths between servers, reducing network congestion.
- If one connection fails, data can still travel through alternative routes.
- It offers excellent fault tolerance and reliability for critical applications.
- It supports large-scale expansion as more sensors, cameras, and servers are added.
- Communication delays are minimized, enabling real-time traffic monitoring and control.

Although Hypercube topology also provides excellent performance, Mesh topology is more practical and easier to implement for large smart city infrastructures.

Part B – Performance Evaluation

Given Data

- Sequential Execution Time (T_s) = 120 seconds
- Parallel Execution Time (T_p) = 20 seconds
- Number of Processors (P) = 8

1. Calculate the Speedup of the Parallel System.

The formula for speedup is:

$$\text{Speedup (S)} = \text{Sequential Execution Time} \div \text{Parallel Execution Time}$$

Substituting the values:

$$S = 120 \div 20$$

$$S = 6$$

Answer

The speedup of the parallel system is **6**.

Interpretation

This means that the parallel system completes the traffic data processing **6 times faster** than the sequential system.

2. Calculate the Efficiency of the System.

The formula for efficiency is:

$$\text{Efficiency (E)} = \text{Speedup} \div \text{Number of Processors}$$

Substituting the values:

$$E = 6 \div 8$$

$$E = 0.75$$

Converting to percentage:

$$\text{Efficiency} = 0.75 \times 100 = 75\%$$

Answer

The efficiency of the system is **75%**.

Interpretation

This indicates that 75% of the available processor resources are effectively utilized, while the remaining 25% is lost due to communication and synchronization overhead.

3. If the Number of Traffic Sensors Doubles in the Future, Explain How Scalability Affects System Performance.

Scalability is the ability of a distributed system to handle increasing workloads by adding more computing resources.

If the number of traffic sensors doubles:

- More traffic data will be generated every second.
- Additional processors or servers can be added to distribute the workload.
- The system can continue processing data efficiently without significant delays.
- Real-time monitoring and traffic signal control remain effective.
- Without scalability, servers may become overloaded, causing slower response times and performance degradation.

Therefore, a scalable system ensures that performance remains stable even as the smart city expands.

4. Explain How Communication Overhead Can Reduce the Performance of Distributed Systems.

Communication overhead refers to the extra time spent exchanging data and synchronizing processes between distributed nodes instead of performing actual computations.

Examples include:

- Sending vehicle count information between servers.
- Sharing congestion reports.
- Synchronizing traffic signal timings.
- Waiting for messages from other processors.

High communication overhead can:

- Reduce system speedup.
- Lower processor efficiency.
- Increase execution time.
- Create network congestion.
- Limit the benefits of parallel processing.

Therefore, minimizing communication overhead is essential for achieving high performance in distributed computing systems.

Part C – Message Passing Interface (MPI)

1. What is MPI and Why is it Important in Distributed Computing?

Message Passing Interface (MPI) is a standardized communication protocol that enables multiple processes running on different computers or processors to exchange information efficiently in a distributed computing environment.

MPI is important because it:

- Enables communication between distributed processors.
- Supports parallel execution of large computational tasks.
- Improves scalability and performance.
- Is portable across different hardware platforms.
- Is widely used in scientific computing, simulations, and real-time data processing.

In the Smart City Traffic Management System, MPI allows distributed servers to exchange congestion data, accident reports, and signal timing information quickly and efficiently.

2. Explain the Role of the Following MPI Functions

a) MPI_Init()

- Initializes the MPI execution environment.
- Must be called before any MPI communication occurs.
- Starts all participating processes.

Purpose: It prepares the distributed program for execution.

b) MPI_Send()

- Sends data from one process to another.
- Requires the destination process ID and message data.

Purpose: It transfers traffic information such as congestion levels or vehicle counts between processors.

c) MPI_Recv()

- Receives data sent by another process.
- Waits until the specified message arrives.

Purpose: It allows processors to receive updated traffic information from other nodes.

d) MPI_Finalize()

- Terminates the MPI execution environment.
- Releases resources used during communication.

Purpose: It safely ends the distributed program.

3. Write a Simple MPI-Based Algorithm Showing How Two Processors Exchange Traffic Congestion Data.

Pseudocode

Start

Call MPI_Init()

If Process ID = 0 Then

 Set Congestion = "Heavy Traffic at Junction A"

 Call MPI_Send(Congestion, Process 1)

End If

If Process ID = 1 Then

 Call MPI_Recv(Congestion, Process 0)

 Display Congestion

End If

Call MPI_Finalize()

Stop

Explanation

Processor 0 detects heavy traffic at a junction and sends the congestion information to Processor 1 using MPI_Send(). Processor 1 receives the message using MPI_Recv() and displays or processes the information. Finally, both processors terminate the MPI environment using MPI_Finalize().

Part D – Critical Analysis

1. Suppose One Server Fails During Traffic Processing. How Can Distributed Systems Maintain Fault Tolerance?

Distributed systems can maintain fault tolerance using several techniques:

Replication

Important traffic data is stored on multiple servers. If one server fails, another copy is immediately available.

Checkpointing

The system periodically saves its current state so processing can resume from the latest checkpoint instead of starting over.

Automatic Failover

Backup servers automatically replace failed servers without interrupting service.

Load Balancing

The workload of the failed server is redistributed among the remaining servers.

Redundant Communication Paths

Alternative network routes allow communication even if one connection fails.

These mechanisms ensure continuous operation of the Smart City Traffic Management System.

2. Which Interconnection Topology Would Provide Better Reliability in This Case?

Mesh Topology provides the highest reliability.

Reasons

- Multiple communication paths exist between nodes.
- Failure of one server or connection does not disconnect the network.
- Data can be rerouted through alternative routes.
- The system continues functioning even during hardware failures.

Therefore, Mesh topology is the best choice for maintaining uninterrupted traffic management services.

3. Suggest Two Techniques to Optimize Communication Between Distributed Nodes in MPI-Based Systems.

Technique 1: Non-Blocking Communication

Functions such as `MPI_Isend()` and `MPI_Irecv()` allow processors to continue computation while communication occurs in the background.

Benefits:

- Reduces waiting time.
- Improves processor utilization.
- Increases overall system performance.

Technique 2: Message Aggregation

Instead of sending many small messages, several updates are combined into one larger message before transmission.

Benefits:

- Reduces communication overhead.
- Lowers network congestion.
- Improves bandwidth utilization.
- Decreases latency.

These optimization techniques significantly improve the efficiency of MPI-based distributed systems.

Conclusion

The Smart City Traffic Management System requires a powerful parallel and distributed computing infrastructure to process massive amounts of real-time traffic data efficiently. Among the evaluated interconnection networks, Mesh topology provides the best balance of scalability, reliability, and fault tolerance. Performance evaluation shows that parallel computing greatly improves execution speed, achieving a speedup of 6 and an efficiency of 75%. MPI enables distributed servers to exchange information effectively through message passing, while fault tolerance techniques and communication optimizations ensure continuous and reliable operation even during server failures or future system expansion. Overall, combining parallel processing, robust interconnection networks, and MPI communication creates an efficient and scalable solution for modern smart city traffic management.